Dynamics, the Action Principle, and the Kinematics That Supports Them

From Quaternions to the Einstein–Yang–Mills Dumbbell, Verified in Lean 4

Ryojiro Kurokawa

Department of Electrical Engineering, Graduate School of Science and Technology
Tokyo University of Science

Preliminary draft, 12 November 2026

Preface

This book is organised around one sentence: *a shape is kinematics; putting in an action produces dynamics.* Every chapter first shows a shape, then states the action, then derives the equations, and finally points to a theorem that has been checked by the Lean 4 proof assistant.

The shape that runs through the book is the *dumbbell*: two round caps joined by a cylindrical neck. With the Einstein–Yang–Mills action the caps become closed radiation universes and the neck becomes the throat of an extremal black hole, $\mathrm{AdS}_2 \times S^2$. Both have the same size ℓ, fixed by the gauge coupling through $\ell^2 = 4\pi G/g^2$. Before we get there we build everything we need from quaternions: rotations, spin, the three-sphere, Lorentz transformations, and the geometry of relativistic momentum.

The intended reader is an undergraduate in the second to fourth year who knows linear algebra, multivariable calculus and the first courses on mechanics and electromagnetism. General relativity, quantum field theory and group theory are not assumed; we build the parts we use.

How to read the status tags

Every result carries one of the following tags, taken directly from the claim ledger of the Lean repository.

proved in Lean	checked in Lean from its stated hypotheses
external theorem	uses a theorem from the literature, cited in the text
numerical	a numerical computation, not checked by Lean
negative result	an idea that was tried and does not work
open	not solved here (and in most cases not solved anywhere)

Physical readings ("this is a gravitational wave") are kept separate from theorems and are called interpretations. Exercises stay within what Lean has checked; each exercise names the theorem that answers it.

A guide for the specialist

This section is written for readers who already know general relativity and gauge theory and want to know, in two pages, what is in this book that they have not seen before, and how far it can be trusted.

What has been checked

The book is a companion to the Lean 4 repository *Cyclic Fueter Theory* (Zenodo concept DOI 10.5281/zenodo.23448911, version 0.6.0). At that version the repository contains 1873 theorem declarations in 171 modules. An axiom audit of 2421 entries (all public theorems and the index aliases) found no `sorry`; 2396 entries depend only on Lean's standard axioms `propext`, `Classical.choice` and `Quot.sound`, and the rest on a subset of them. Lean verifies each theorem from its declared hypotheses. It does not verify that the hypotheses describe nature, and results quoted from the literature are tagged external theorem.

Claims of novelty

The following statements are, to the author's knowledge, not stated in this form in the literature the author has consulted. The claim is deliberately weak: no systematic review has been carried out, no specialist has confirmed it, and several items are special cases or rearrangements of known results. They are listed so that they can be checked.

(1) **Linear rigidity** (Chapter 19). Every linear perturbation of the Einstein–Yang–Mills Tolman universe that preserves spherical symmetry, the magnetic ansatz and conformal flatness is generated by a conformal Killing vector of the Einstein static universe, so it is pure gauge.

(2) **Shape of the dumbbell** (Chapter 16). Time-symmetric, conformally flat slices satisfy the Bogomolny-type inequality $r'^2 \le W^2$. The case $W = \pm r'$ saturates it and gives $r^2 + \ell^2 r'^2 = \ell^2$. The cap is the regular solution of this equation and the Bertotti–Robinson neck is its singular solution, the envelope of the caps. The junction is fixed by continuity alone, is C^1 and never C^2, and the Hamiltonian constraint admits exactly the two values 0 and $-1/\ell$ of r'' there.

(3) **Null junction** (Chapters 16–17). A timelike junction between cap and neck is impossible, and the jump $[W_u] = 1/(\sqrt{2}\,\ell)$ is transported unchanged along the null fronts.

(4) **Weyl invariant** (Chapter 16). In this sector the Weyl invariant equals $3\big(2m/r - \ell^2(1 - W^2)^2/r^2\big)$, so conformal flatness says that the Misner–Sharp mass equals the magnetic self-energy on every sphere. This is a special case of the known relation between Weyl curvature and the Misner–Sharp mass.

(5) **One pair of constraints** (Chapter 23). The same Hamiltonian and momentum constraints reduce to the Friedmann constraint on the cap and to a flat two-dimensional Klein–Gordon equation on the Kantowski–Sachs side, where the neck $r = \ell$ is a fold of minisuperspace, invariant under $r \mapsto \ell^2/r$. The neck length is canonically conjugate to $-r\dot{r}$. Minisuperspace quantisation of Reissner–Nordström interiors is known; the reading in terms of the dumbbell is what is claimed.

(6) **Constant-area data** (Chapter 26). Time-symmetric data of constant areal radius r_0 obey $W'^2 = (1 - \ell^2(1 - W^2)^2/r_0^2)/(2\ell^2)$, with the lump $W = \sqrt{2}/\cosh(x/\ell)$ at $r_0 = \ell$. Only the neck keeps its area at the next instant.

(7) **Weak gravity conjecture** (Chapter 26). With an adjoint Higgs field the electric condition for the W boson and the magnetic condition for the BPS monopole coincide, $4\pi G v^2 \leq 1$. Together with the stability bound of the neck they give $3/4 \leq (m_W \ell)^2 \leq 1$, and the neck exists only for $\Lambda \leq g^2/(16\pi G)$.

(8) **Formalisation** (whole book). The spherically symmetric Einstein–Yang–Mills–Dirac sector is machine-checked from one reduced action through the constraints, the known limits and the thermodynamic relations.

What is not claimed

The cap and neck solutions themselves, the reduced action, the canonical form and the mass function, JT gravity near the neck, black-hole thermodynamics, the sphaleron zero mode, and the weak gravity and Festina Lente conjectures are standard. The general solution of the Wheeler–DeWitt equation with matter, the quantisation of gravitational waves beyond the linear level, and any derivation of the fine-structure constant are *not* obtained; Chapter 27 lists them as open problems.

Acknowledgements

The Lean proofs, numerical checks and parts of the text were prepared with the assistance of a large language model (Claude, Anthropic); the author has checked the results and takes responsibility for the content.

Contents

Preface **ii**

A guide for the specialist **iii**

Plan of the book **1**

16 The shape of the dumbbell **3**
16.1 Kinematics: what a slice looks like 3
16.2 The action and the constraint 4
16.3 A Bogomolny inequality 4
16.4 The circle and its envelope 5
16.5 Theorem A: the junction is forced 6

Plan of the book

This preliminary draft contains the front matter and Chapter 16, which fixes the style of the whole book. The other chapters are planned as follows; the right-hand column names the main Lean modules each chapter draws on.

	Title	Main Lean modules	Status
Part I. Quaternions and rotations			
1	From complex numbers to quaternions	`AlphaI`	planned
2	Rotations and the three-sphere	`AlphaI`, `AlphaCompleteB`	planned
3	Fueter operators	`AlphaI_Operators`, `AlphaI_C`, `AlphaI_CFamily`	planned
4	Lorentz transformations: rotation $\times$ boost	`AlphaPillar_Polar`, `AlphaRoot_Duality`	planned
Part II. The quantum sphere and the mass shell			
5	The mass-shell hyperboloid	`AlphaB_QH`, `AlphaI_H3`	planned
6	The quantum sphere	`AlphaI_QBij`	planned
7	The Dirac equation and its projector	`AlphaV`, `AlphaV_FWOrder`	planned
8	Connections and curvature: Fueter, Berry, Wigner	`AlphaI_AW`, `AlphaV_Wigner`	planned
9	Monopoles and zero modes	`AlphaCompleteC`, `AlphaCompleteD`, `ZeroModeL2`	planned
Interlude A. The Standard Model from an internal sphere			
A	$SO(10)$, Pati–Salam, $B-L$ and chirality	`AlphaVII_*`, `AlphaFix_Gauge`, `AlphaFix_Z28`	planned
Part III. Curved space and field equations			
10	Moving frames and curvature	`Frame`, `AlphaS4_Curv`	planned
11	Slices and constraints	`AlphaII`, `AlphaIII_Codazzi`	planned
12	Gauge fields	`GaugeCovariance`, `AlphaIX_YM`, `AlphaVII_Maxwell`	planned
13	The action principle	`AlphaS1_Derivation`, `AlphaS2_*`, `AlphaIX_Classical`	planned
Part IV. The dumbbell			
14	Three phases	`AlphaII_*`, `AlphaFix_Geo`, `C2`	planned
15	Cap and neck	`AlphaII_EYM`, `AlphaIII_Glue`	planned
16	The shape of the dumbbell	`AlphaDumbbell_Shape`, `AlphaDumbbell_Bogomolny`, `AlphaJunction_Unique`	**drafted**
17	Junctions	`AlphaIII*`, `AlphaS4_*`, `AlphaReal_Junction`	planned
18	Time evolution	`AlphaIV*`, `AlphaS5_*`, `AlphaReal_Rest`	planned
19	The conformal frame and Ω	`AlphaPillar_Conformal`, `AlphaOmega_*`	planned
20	Stability	`AlphaXI_*`, `AlphaPillar_Smooth`	planned
Interlude B. Contact with string theory			

	Title	Main Lean modules	Status
B	WZW, the standard embedding and attractors	`AlphaVIII_*`	planned
Part V. Heat and quanta			
21	Black-hole thermodynamics	`AlphaIX_Thermo`, `AlphaSphericalClose`	planned
22	Quantum mechanics of the neck	`AlphaIX_JT`, `AlphaFamous`	planned
23	Quantum cosmology	`AlphaCanonical`, `AlphaQuantumSpherical`, `AlphaKS_WDW`	planned
24	Quantum fields in curved spacetime	`AlphaV_FRW`, `AlphaV_Anomaly`	planned
25	Fermions	`AlphaFermion`, `AlphaV_S3Dirac`	planned
Part VI. Shape, swampland and open problems			
26	Shape degrees of freedom and the weak gravity conjecture	`AlphaGeneral`, `AlphaConstArea`, `AlphaWGC`	planned
27	Open problems	`AlphaIdeas_Check`, ledger	planned

Chapter 16

The shape of the dumbbell

The question of this chapter

Why can the cap and the neck not be joined smoothly? *Answer:* because the neck is the envelope of the caps. Both solve the same first-order equation; the cap is its regular solution and the neck its singular one, and a curve that switches from a solution to the envelope of the solutions is C^1 but never C^2.

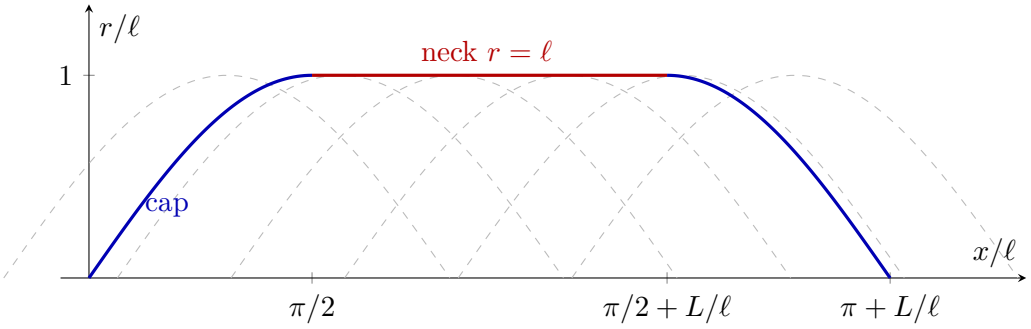


Figure 16.1: The areal radius along a time-symmetric slice of the dumbbell (solid). The dashed curves are other members of the family $r = \ell \sin((x-c)/\ell)$; the neck $r = \ell$ touches every one of them at its top. It is their envelope.

16.1 Kinematics: what a slice looks like

Take a moment of time symmetry and look at the spatial slice. Spherical symmetry lets us write its metric as

$$ds^2 = dx^2 + r(x)^2\, d\Omega^2, \tag{16.1}$$

where x is proper distance along the axis and $4\pi r^2$ is the area of the sphere at x; r is called the *areal radius*. The magnetic $SU(2)$ Yang–Mills field compatible with this symmetry is described by one function $W(x)$ (Chapter 12): its radial magnetic field is $(1-W^2)/r^2$ and its tangential one is W'/r.

Two numbers attached to each sphere will organise the whole chapter.

Definition 16.1. The *Misner–Sharp mass* of the sphere at x on a time-symmetric slice is $m = \frac{r}{2}(1 - r'^2)$. The *magnetic self-energy* of that sphere is $\ell^2(1-W^2)^2/(2r)$.

For the cap $r = \ell \sin(x/\ell)$ one finds $m = \frac{\ell}{2}\sin^3(x/\ell)$, and for the neck $r = \ell$ one finds $m = \ell/2$.

The slice is *conformally flat* when its Weyl curvature vanishes. In this sector the relevant invariant can be computed in double null coordinates, and it comes out as a difference of the two numbers above.

Proposition 16.2 (Weyl curvature = mass − self-energy). proved in Lean *In spherically symmetric Einstein–Yang–Mills theory the conformally invariant combination of the metric functions equals*

$$3\Big(\frac{2m}{r} - \frac{\ell^2(1-W^2)^2}{r^2}\Big).$$

Hence the slice is conformally flat exactly when $2mr = \ell^2(1-W^2)^2$: on every sphere the mass equals the magnetic self-energy.

Lean: `CFT.DS.ds_weyl_mass`.

16.2 The action and the constraint

The Einstein–Yang–Mills action of Chapter 13, reduced to spherical symmetry, gives one equation that every slice must satisfy, the Hamiltonian constraint. At a moment of time symmetry it reads

$$-4rr'' + 2(1 - r'^2) = 4\ell^2\Big(W'^2 + \frac{(1-W^2)^2}{2r^2}\Big), \qquad \ell^2 = \frac{4\pi G}{g^2}. \tag{16.2}$$

The left side is r^2 times the scalar curvature of the slice; the right side is r^2 times $16\pi G$ times the Yang–Mills energy density. You can check by hand that the cap $r = \ell\sin(x/\ell)$, $W = \cos(x/\ell)$ and the neck $r = \ell$, $W = 0$ both satisfy it: in both cases the two sides are $6\sin^2(x/\ell)$ and 2.

Multiplying (16.2) by $r'/4$ turns it into a statement about the mass:

$$m' = r'\,\ell^2\Big(W'^2 + \frac{(1-W^2)^2}{2r^2}\Big). \tag{16.3}$$

Lean: `CFT.DS.ds2_mass_prime`. The mass grows wherever the sphere grows, at a rate fixed by the energy density.

16.3 A Bogomolny inequality

Now impose conformal flatness, $2mr = \ell^2(1-W^2)^2$, and differentiate it along the slice. Using (16.3) and dividing by $2\ell^2$ gives

$$rr'W'^2 + 2WW'(1-W^2) + \frac{(1-W^2)^2 r'}{r} = 0. \tag{16.4}$$

This looks unpromising, but multiplied by r' it is a sum of squares.

Lemma 16.3 (completing the square). proved in Lean *For all real $r \neq 0$, r', W, W',*

$$r'\Big(rr'W'^2 + 2WW'(1-W^2) + \frac{(1-W^2)^2 r'}{r}\Big) = \frac{\big(rr'W' + W(1-W^2)\big)^2 + (1-W^2)^2\,(r'^2 - W^2)}{r}.$$

Lean: ds2_bogomolny

```
theorem ds2_bogomolny (r r' W W' : ℝ) (hr : r ≠ 0) :
    r' * (r * r' * W' ^ 2 + 2 * W * W' * (1 - W ^ 2)
          + (1 - W ^ 2) ^ 2 * r' / r)
      = ((r * r' * W' + W * (1 - W ^ 2)) ^ 2
          + (1 - W ^ 2) ^ 2 * (r' ^ 2 - W ^ 2)) / r := by
  field_simp
  ring
```

Theorem 16.4 (Bogomolny inequality for conformally flat slices)**.** proved in Lean *On a time-symmetric, conformally flat slice of spherically symmetric Einstein–Yang–Mills theory, at every point where $W^2 \neq 1$,*

$$r'^2 \leq W^2.$$

Equality holds if and only if $W = \pm r'$, and then the slice obeys the first-order equation $rr'W' = -W(1 - W^2)$.

Proof. By (16.4) the left side of the lemma vanishes, so the numerator on the right vanishes. It is a square plus $(1-W^2)^2(r'^2 - W^2)$ with $(1-W^2)^2 > 0$. If $r'^2 > W^2$ the numerator would be positive. At equality the second term is zero, so the square is zero. □

Lean: `CFT.DS.ds2_num`, `ds2_W_ge`, `ds2_saturated`.

Remark. Earlier chapters imposed $W = r'$ by hand, motivated by the observation that the Yang–Mills potential can be identified with the spin connection of the slice (Chapter 15). Theorem 16.4 shows that this condition is not an extra assumption inside the conformally flat class: it is the case of equality of an inequality that every such slice satisfies. This is the same logic as the Bogomolny bound for monopoles, where the first-order equations describe the configurations that saturate an energy inequality.

16.4 The circle and its envelope

In the saturating case $W = r'$ we have $m = \frac{r}{2}(1 - W^2)$, and conformal flatness $2mr = \ell^2(1 - W^2)^2$ becomes $r^2(1 - W^2) = \ell^2(1 - W^2)^2$. Dividing by $1 - W^2$:

Theorem 16.5 (the shape equation)**.** proved in Lean *On a time-symmetric, conformally flat slice with $W = r'$ and $W^2 \neq 1$,*

$$r^2 + \ell^2 r'^2 = \ell^2. \tag{16.5}$$

Lean: `CFT.DS.ds_circle_of_flat`.

Equation (16.5) says that the point $(r, \ell W) = (r, \ell r')$ moves on a circle of radius ℓ (Figure 16.2). The cap goes round half of it, from the north pole $(0, \ell)$ through the equator $(\ell, 0)$ to the south pole $(0, -\ell)$. The neck does not move at all: it sits at the single point $(\ell, 0)$.

Lean: ds_cap_on_circle

```
theorem ds_cap_on_circle (l x : ℝ) :
    (l * sin (x / l)) ^ 2 + l ^ 2 * cos (x / l) ^ 2 = l ^ 2 := by
  linear_combination l ^ 2 * sin_sq_add_cos_sq (x / l)
```

Equation (16.5) is of first order, and first-order equations can have solutions that are not members of the general family. Differentiating it gives $2r'(r + \ell^2 r'') = 0$, so at every point one of two things happens.

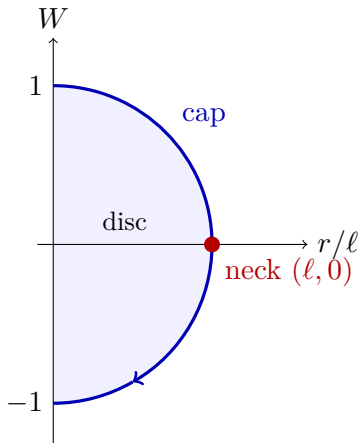


Figure 16.2: The (r, W) plane. Conformally flat slices with $W = r'$ live on the circle $r^2 + \ell^2 W^2 = \ell^2$. The cap runs along it; the neck is the point $(\ell, 0)$. The shaded disc is the region where conformally flat solutions have real null directions (Proposition 16.8).

Theorem 16.6 (two branches). proved in Lean *Let r be twice differentiable and satisfy* (16.5). *Then at every point*

$$r' = 0 \qquad \text{or} \qquad r'' = -\frac{r}{\ell^2}.$$

Lean: `CFT.DS.ds_branch`.

The second branch is the harmonic equation; its solutions are the shifted caps $r = \ell \sin((x - c)/\ell)$, the regular solutions. The first branch, with $r^2 = \ell^2$, is the constant $r = \ell$: the neck. It is the *singular solution* of (16.5), the envelope of the family of caps (Figure 16.1). At $r = \ell$ the equation $r' = \pm\sqrt{\ell^2 - r^2}/\ell$ loses uniqueness, and this is the only place where a solution can leave the family of caps.

Corollary 16.7. *A C^1 curve that follows a cap and then the neck must switch at a point where the cap reaches $r = \ell$. There the first derivatives agree ($r' = 0$ on both sides) but the second derivatives are $-1/\ell$ and 0. The dumbbell is C^1 and never C^2.*

This is the geometric content of the "jump" that appeared in earlier chapters. It is not a defect put in by hand; it is the standard relation between a family of curves and its envelope: an envelope touches each curve, so slopes agree, but it does not bend with it.

Proposition 16.8 (the disc). proved in Lean *For conformally flat solutions the slope dW/dr along a null direction solves a quadratic equation with coefficients $(1 - W^2)^2/r$, $2W(1 - W^2)$ and $(r^2 - \ell^2(1 - W^2)^2)/r$. Its discriminant is*

$$\frac{4(1 - W^2)^3}{r^2}\big(\ell^2(1 - W^2) - r^2\big),$$

so for $W^2 < 1$ real slopes exist exactly in the disc $r^2 + \ell^2 W^2 \le \ell^2$. The two roots coincide on the circle. The whole cap spacetime, $r = \ell \cos\eta \sin\chi$ and $W = \cos\chi$, lies in the disc, since $r^2 + \ell^2 W^2 = \ell^2 - \ell^2 \sin^2\eta \sin^2\chi$.

Lean: `CFT.DS.ds_slope_disc`, `ds_slope_real`, `ds_cap_disc`.

16.5 Theorem A: the junction is forced

Collecting the pieces, the way the cap meets the neck admits no choice.

Theorem 16.9 (uniqueness of the junction). proved in Lean *Consider the cap $r = \ell \sin(x/\ell)$ and the neck $r = \ell$ with $W = r'$.*

(i) The areal radius is continuous only at the equator $x = \pi\ell/2$ (`ju_location`).

(ii) There $r' = \cos(x/\ell) = 0$, so the first derivatives agree automatically (`ju_C1_auto`).

(iii) The second derivatives are $-1/\ell$ on the cap (`ju_cap_second`) and 0 on the neck, and at $r = \ell$, $r' = 0$ the Hamiltonian constraint (16.2) reduces to $r''(r'' + 1/\ell) = 0$, whose only roots are these two values (`rr_two_roots`).

(iv) The junction cannot be a timelike tube: along it the cap's equatorial radius $\sqrt{\ell^2 - \tau^2}$ differs from ℓ for $\tau \neq 0$ (`rj_tube_mismatch`).

(v) Along a null front the jump $[W_u]$ keeps its value at the corner, $1/(\sqrt{2}\,\ell)$ (`ju_jump_fixed`).

The combined statement is `CFT.JU.ju_theorem_A`.

Part (i) deserves a comment. One might expect the location of the junction to be a choice. It is not: the largest radius of the cap is exactly the radius of the neck, so they can touch only at the top of the cap, and there the slopes agree by themselves. "Glue continuously" already implies "glue in a C^1 way".

Where this stands

Known. The cap (the closed radiation universe driven by the homogeneous Yang–Mills field) and the neck (the magnetic Bertotti–Robinson spacetime) are classical solutions [1, 2, 3]. That the throat of an extremal black hole has a free length, and that it saturates an area–charge inequality, is known for Einstein–Maxwell theory [5, 6]. The relation between the Weyl curvature and the Misner–Sharp mass is known in general [4].

Claimed as new, in the weak sense of the guide for the specialist. The Bogomolny-type inequality $r'^2 \leq W^2$ for conformally flat slices (Theorem 16.4), the reading of the neck as the singular solution and envelope of the caps (Theorems 16.5 and 16.6), and the statement that the junction is forced (Theorem 16.9).

Assumptions to keep in mind. Spherical symmetry, a purely magnetic field, time symmetry and conformal flatness. Outside the class $W = r'$ the two roots of (iii) disappear, so these theorems do not exclude a smooth interpolation in general. Inside the conformally flat class a numerical search for a smooth static interpolation failed negative result (Chapter 19).

Open. Why the dumbbell is glued at all is not answered by these theorems; they say only how it must be glued if it is. Chapter 26 shows that a fixed total volume $2\pi^2\ell^3 + 4\pi\ell^2 L$ fixes the neck length L, and Chapter 23 shows that quantum mechanically L is conjugate to $-r\dot{r}$. open

Exercises

Each exercise is answered by the named Lean theorem. Read its statement in the repository and compare it with your own answer.

Exercise 16.1. Check by hand that the cap $r = \ell \sin(x/\ell)$, $W = \cos(x/\ell)$ lies on the circle (16.5), and that the neck $(r, W) = (\ell, 0)$ lies on it too. *Answer:* `ds_cap_on_circle`, `ds_neck_on_circle`.

Exercise 16.2. Multiply the Hamiltonian constraint (16.2) by $r'/4$ and show that it becomes (16.3). *Answer:* `ds2_mass_prime`.

Exercise 16.3. Expand both sides of the completing-the-square lemma and check that they agree. Where is $r \neq 0$ used? *Answer:* `ds2_bogomolny`.

Exercise 16.4. Theorem 16.5 assumes $W^2 \neq 1$. Find what goes wrong in the proof when $W^2 = 1$, and describe the slices with $W \equiv \pm 1$. *Answer:* the division in `ds_circle_of_flat`; for the slices, `fam_birkhoff_u` in Chapter 21.

Exercise 16.5. Show that $\ell \sin(x/\ell) = \ell$ with $0 < x < \pi\ell$ forces $x = \pi\ell/2$. *Answer:* `ju_location`.

Exercise 16.6. Compute the derivative of $\cos(x/\ell)$ at $x = \pi\ell/2$. *Answer:* `ju_cap_second`.

Exercise 16.7. Solve $2\ell^2 y^2 + 2\ell y = 0$ and explain why its two roots are the second derivatives of the neck and of the cap at the junction. *Answer:* `rr_two_roots`.

Exercise 16.8. Show that for $W^2 < 1$ the discriminant of Proposition 16.8 is non-negative if and only if $r^2 + \ell^2 W^2 \leq \ell^2$. *Answer:* `ds_slope_real`.

Bibliography

[1] R. C. Tolman, *Relativity, Thermodynamics and Cosmology* (Oxford University Press, 1934).

[2] B. Bertotti, Uniform electromagnetic field in the theory of general relativity, Phys. Rev. **116** (1959) 1331.

[3] I. Robinson, A solution of the Maxwell–Einstein equations, Bull. Acad. Polon. Sci. **7** (1959) 351.

[4] C. W. Misner and D. H. Sharp, Relativistic equations for adiabatic, spherically symmetric gravitational collapse, Phys. Rev. **136** (1964) B571.

[5] S. Dain, J. L. Jaramillo and M. Reiris, Area–charge inequality for black holes, Class. Quantum Grav. **29** (2012) 035013.

[6] Area–charge inequalities and rigidity of time-symmetric initial data sets, arXiv:2507.13040 (2025).